



CENTRE FOR INTERNATIONAL
EXPERIENCE

CIE Resource Hub Conversational Agent

An AI-powered assistant for international student support

Team 4 · RSM 8430 Final Project

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The Problem

- ① CIE serves thousands of international students
- ② Information is scattered across 80+ web pages
- ③ Students need personalized, multi-step guidance
- ④ Incorrect information about visas or UHIP can have serious consequences



The Dataset

Source

CIE Resource & Information Hub
(internationalexperience.utoronto.ca)

Coverage

Pre-Arrival, UHIP, Immigration, Finances, Events, Peer Support

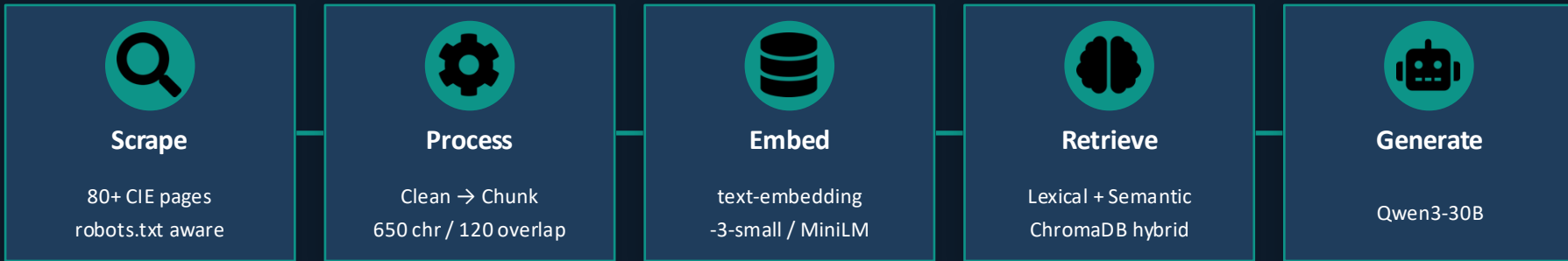
Format

Semi-structured HTML

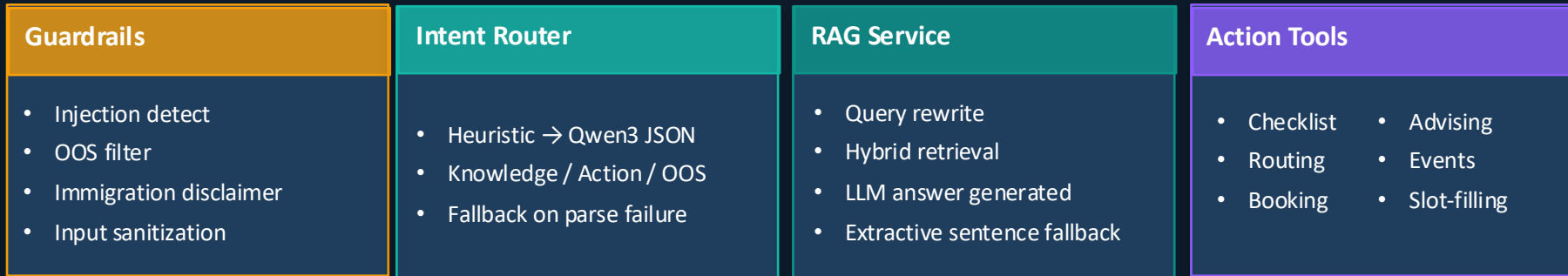
Fallback

International Student Handbook PDF + archived pages

OFFLINE PIPELINE



RUNTIME AGENT COMPONENTS



Evaluation Results



UNIVERSITY OF
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CENTRE FOR INTERNATIONAL
EXPERIENCE

71.9%

Overall Pass Rate

23 / 32 cases

6 categories

- Knowledge
- Action
- Multi-turn
- Out-of-scope
- Guardrails
- Error handling

Intent Accuracy

100%

Action Success Rate

100%

Safe Response

100%

Context Precision

93.8%

Error Handling

93.8%

Answer Relevancy

90.6%

Faithfulness

83.2%



Key Failure Modes

- ① Slot-filling order confusion in multi-turn booking/advising flows (3 cases)
- ② Faithfulness metric penalizes memory-recall answers that use session history instead of RAG sources (6 cases)

Learnings & Limitations

✓ What Worked Well

- ✓ Hybrid search wins on specialist vocabulary
- ✓ Three-tier fallback - stable, honest, no hallucinations
- ✓ Templates > LLM for structured outputs
- ✓ State machine > prompt hacks for multi-turn
- ✓ 100% intent accuracy - dual-classifier validated

⚠ Challenges & Known Limits

- ! Slot-filling order: fragile at 6 parameters
- ! Faithfulness metric penalizes memory-recall unfairly
- ! Hardcoded guardrails miss novel out-of-scope inputs
- ! Knowledge freshness requires manual re-scraping
- ! Session state is local and doesn't follow the user

Thank you